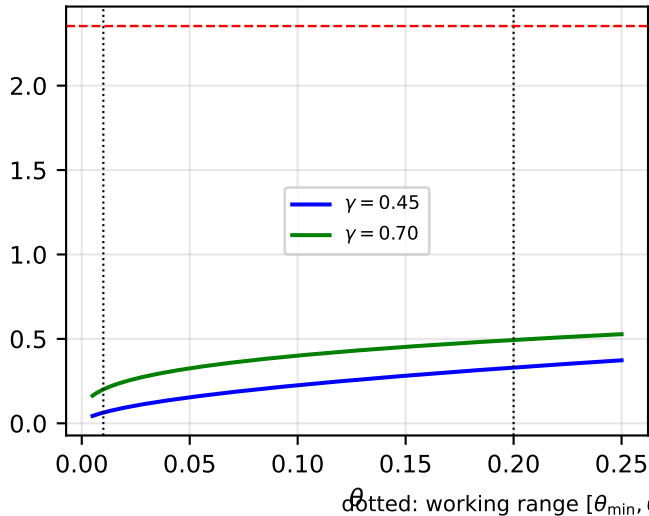


$\theta\phi(\theta)$: sup at $\theta_{\max}$ always



$\theta\phi(\theta)^2$: sup at endpoint depending on $\gamma \lesssim \frac{1}{2}$

